

How the project was built — what we did, in order

Each phase had a gate that had to pass before the next one started. Nothing was tuned after results were seen.

1	Data layer Load 5 drift streams in original time order, never shuffled.	GATE · all five loaders return correctly shaped arrays datasets.py · elec2, insects ×3, covtype
2	Cost accounting work_units = passes × rows, read from the fitted model.	GATE · a 300-tree fit costs exactly 30× a 10-tree fit accounting.py · energy logged, never decisive
3	Repair mechanisms Skip, nudge and rebuild for four model families.	GATE · nudge cheaper than rebuild; forest size constant mechanisms.py · 4% · 0.6–1% · 80%
4	The selector Try the cheap option, verify on recent rows, escalate if it fails.	GATE · holdout rows never appear in training data selector.py · the contribution
5	Baseline policies Four competitors with identical actions.	GATE · all five policies run end to end policies.py · FixedSchedule(k=3) is the one to beat
6	Experiment runner Stream, detect, adapt, log every run and every alarm.	GATE · full grid completes — 500 runs, 0 failures runner.py · 5 × 4 × 5 × 5
7	Analysis Friedman, Nemenyi, Wilcoxon + Holm, Pareto, 11 figures.	GATE · determinism verified before collapsing to n = 60 analysis.py · n = 20 kept as a stricter check
7b	Second detector — DDM Repeat all 500 runs with a one-sided detector.	RESULT · it falsified our own explanation, and we kept it ddm_contrast.py · replay reproduced exactly
8	Packaging Ship the selector; README states the negative results first.	GATE · clean install; 251 tests pass against the package package/ · 0.1.0 · defects documented

RUNNING ALONGSIDE EVERY PHASE

251 tests — invariants · exact cost · chunked prediction equals row-by-row · verify_*.py phase demos
diagnostics that answered what the grid could not: nudge_window_diagnostic.py · nudge_overwrite_demo.py · placeholder_sanity.py